

Year 9 Mathematics

Mini-Test – Trigonometry & Scientific Notation

Resource Section (Calculators and notes allowed)



Baldivis
Secondary College

Time: 40 mins

Name: _____

~~MISSIEA BEAGSFORD~~
~~NASHMINE GREAT~~

/ 49 marks

Multi-choice Section

5 marks

1. Which is the correct scientific notation representation of the following number: 47 000

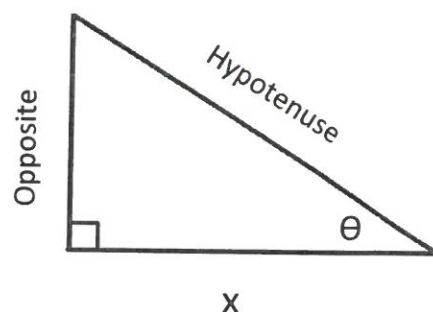
- a. 47.00×10^4
- b. 4.7×10^{-4}
- ☒ c. 4.7×10^4
- d. 0.47×10^4

2. Which is the correct representation of the following number: 9.324×10^{-5}

- a. 9,324,000
- b. 932,400
- c. 0.000009324
- ☒ d. 0.00009324

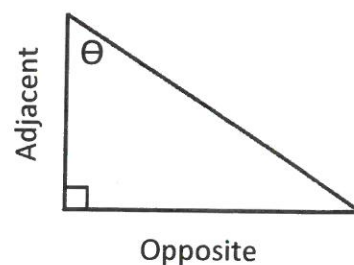
3. What is the correct name for the missing side (x) in the picture:

- a. Hypotenuse
- ☒ b. Adjacent
- c. Opposite
- d. Pythagoras



4. Which one is the correct trigonometry ratio for the following triangle:

- a. $\sin \theta = \frac{\text{Opposite}}{\text{Hypoenuse}}$
- ☒ b. $\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$
- c. $\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$



5. Which one is the correct answer: $\cos 45^\circ = \frac{13}{x}$

- ☒ a. $x = 18.38$
- b. $x = 24.75$
- c. $x = 17.09$
- d. $x = 18.28$

Question 6
12 marks

Complete the following table:

Scientific Notation	Numeral
3×10^1	30
3×10^2	300
3×10^3	3000
3×10^4	30 000

a) Write the following as a numeral.

$2 \times 10^5 = 200000$

$1.9 \times 10^{-1} = 0.19$

$8.3 \times 10^7 = 830000000$

$4 \times 10^{-5} = 0.00004$

b) Write the following numbers in scientific notation.

$60 = 6 \times 10^1$

$420,000,000 = 4.2 \times 10^8$

$0.0000064 = 6.4 \times 10^{-6}$

$0.00005008 = 5.008 \times 10^{-5}$

Question 7
5 marks

 Simplify the following. Write each answer in scientific notation. *Show all your working out.*

a. $\frac{6.6 \times 10^4}{2.99 \times 10^6}$

2.207×10^{-2}

b. $(5.91 \times 10^3)(9 \times 10^4)$

53.19×10^7

5.319×10^8

 c. Find the difference between 4.2×10^7 and 4.2×10^5 .

$$\begin{aligned}
 &42\ 000\ 000 - 420\ 000 \\
 &= 41\ 580\ 000 \\
 &= 4.158 \times 10^7
 \end{aligned}$$

Question 8

4 marks

Find the value of each of the trigonometric ratio, rounding to 3 decimal places

a. $\sin 62^\circ = \underline{0.883}$

b. $\tan 79^\circ = \underline{5.145}$

Find the value of each angle to the nearest degree.

c. $\sin \theta = 0.7547$ $\theta = \underline{49^\circ}$

d. $\cos \theta = 0.5736$ $\theta = \underline{55^\circ}$

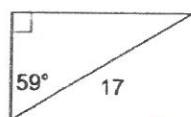
Question 9

9 marks

Solve the following by:

1. Labelling the sides (O, A, H)
2. Choose the correct trigonometry ratio
3. Use the ratio to find the length of the side marked with a letter (to 3 d.p).
4. Show all your working out.

a. m



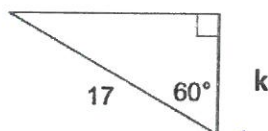
$$\sin \theta = \frac{O}{H}$$

$$\sin 59 = \frac{m}{17}$$

$$m = 17 \times \sin 59$$

$$m = 14.572$$

b.



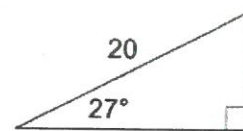
$$\cos \theta = \frac{A}{H}$$

$$\cos 60 = \frac{k}{17}$$

$$k = 17 \times \cos 60$$

$$k = 8.5$$

c.



$$\cos \theta = \frac{A}{H}$$

$$\cos 27 = \frac{y}{20}$$

$$y = 20 \times \cos 27$$

$$y = 17.82$$

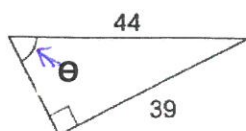
Question 10

6 marks

Solve the following by:

1. Label the sides (O, A, H)
2. Choose the correct trigonometry ratio
3. Use the ratio to find the value of angle marked with a symbol.
4. Show all your working out.

a.

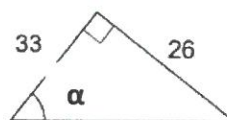


$$\sin \theta = \frac{39}{44}$$

$$\theta = \sin^{-1}(39 \div 44)$$

$$\theta = \underline{62^\circ}$$

b.

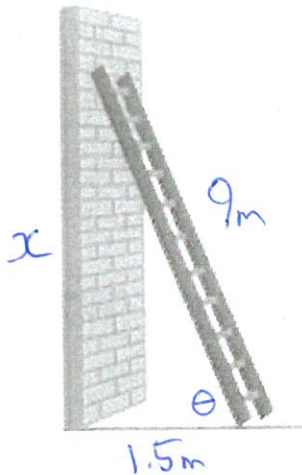


$$\tan a = \frac{O}{A}$$

$$a = \tan^{-1}(26 \div 33)$$

$$a = \underline{38^\circ}$$

- a. A 9 m ladder rests against the side of a wall. The bottom of the ladder is 1.5 m from the base of the wall. Label the diagram to help you. *Show all working out.*
- i. Determine the measure of the angle between the ladder and the ground.



$$\cos \theta = \frac{A}{H}$$

$$\cos \theta = \frac{1.5}{9}$$

$$\theta = \cos^{-1}(1.5 \div 9)$$

$$\theta = 80^\circ$$

Angle formed between ladder and ground is 80°

- ii. How far up the wall does the ladder reach?

$$b^2 = c^2 - a^2$$

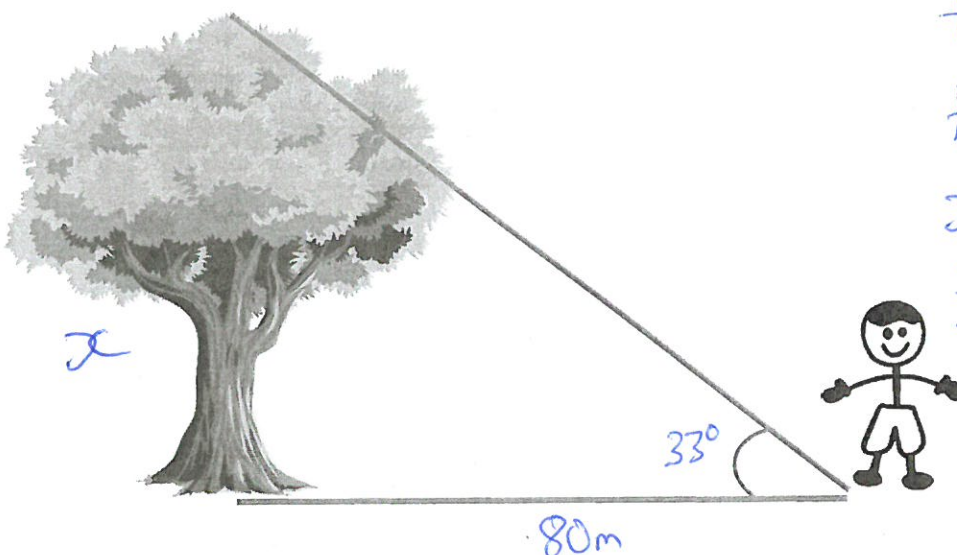
$$x^2 = 9^2 - 1.5^2$$

$$x = \sqrt{81 - 2.25}$$

$$x = 8.87m$$

+ reaches 8.87m up the wall

- b. Craig wants to measure the height of a tree. He walks exactly 80 m away from the tree. The angle from the ground to the top of the tree is 33° . How tall is the tree?
- o Show all your working out



$$\tan \theta = \frac{O}{A}$$

$$\tan 33 = \frac{x}{80}$$

$$x = 80 \times \tan 33$$

$$x = 51.95m$$

The tree is 51.95m tall